Progressive exercises

E1 · Practice. Define on $\mathbb{N}$ the sequences (u_n) and (v_n) by $u_n = \frac{4n-1}{n+2}$ and $v_n = n^2 - 7n$. Determine their limits, justifying the transformations used.

E2 · Practice. Define $u_n = \sqrt{n+1} - \sqrt{n}$ for every $n \in \mathbb{N}$. Using the conjugate, determine the limit of (u_n).

E3 · Practice. Define $u_n = \frac{\cos n}{n}$ for every integer $n \geq 1$. Bound u_n above and below, then deduce the limit of (u_n).

E4 · Practice. The sequence (u_n) is increasing and satisfies $u_n < 5$ for every $n \in \mathbb{N}$. Justify that it converges and state what can be asserted about its limit. Is it necessarily equal to 5? Justify.

E5 · Practice. Define $u_n = 3 - 2(-0.6)^n$ for every $n \in \mathbb{N}$. Determine the limit of (u_n). Is this sequence monotonic? Justify.

Textbook or past-paper work

Barbazo, p. 62, exercises 17 and 24. In S16: exercise 17, to resolve an indeterminate form by dividing by a power of n. At home after S20: exercise 24, to write up an application of the squeeze theorem.